Biology Study Material

Chapter 1: Cell Structure and Function

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1. Introduction to Cells

The cell is the basic structural, functional and biological unit of life.

Every living organism is made up of one or more cells.

The branch of biology that studies cells is called Cytology.

Cells perform:

- Respiration
- Nutrition
- Growth
- Reproduction
- Excretion
- Transportation

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2. Cell Theory

Scientists:

- Matthias Schleiden
- Theodor Schwann

- Rudolf Virchow

Cell theory states:

1. All living organisms are composed of cells.
2. The cell is the basic unit of life.
3. New cells arise from existing cells.

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3. Types of Cells

Prokaryotic Cells

Characteristics:

- Small
- Simple
- No nucleus
- No membrane-bound organelles

Examples:

- Bacteria
- Archaea

Eukaryotic Cells

Characteristics:

- Large
- Complex
- Have a nucleus
- Have membrane-bound organelles

Examples:

- Plants
- Animals
- Fungi
- Protists

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4. Cell Structure

Major components:

- Cell membrane
- Cytoplasm
- Nucleus
- Organelles

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5. Cell Membrane

Functions:

- Protects the cell
- Regulates movement
- Maintains internal balance

Properties:

- Flexible
- Selectively permeable

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6. Cytoplasm

Functions:

- Holds organelles
- Supports chemical reactions

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7. Nucleus

Functions:

- Controls activities

- Stores DNA

Components:

- Nuclear membrane
- Nucleoplasm
- Nucleolus
- Chromosomes

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8. Mitochondria

Known as:

Powerhouse of the cell

Functions:

- Produce ATP
- Cellular respiration

Characteristics:

- Own DNA
- Self replication

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9. Ribosomes

Function:

Protein synthesis

Types:

- Free ribosomes
- Bound ribosomes

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10. Endoplasmic Reticulum

Rough ER

- Ribosomes attached
- Protein synthesis

Smooth ER

- No ribosomes
- Lipid synthesis

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11. Golgi Apparatus

Functions:

- Modifies proteins
- Packages proteins
- Transports proteins

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12. Lysosomes

Functions:

- Waste digestion
- Destroy damaged organelles

Nickname:

Suicidal bags of the cell

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13. Vacuoles

Functions:

- Store water

- Store nutrients
- Store waste

Plant cells have larger vacuoles.

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14. Chloroplast

Present in:

Plant cells only

Functions:

- Photosynthesis

Contains:

Chlorophyll

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15. Plant Cell vs Animal Cell

Plant Cell:

- Cell wall

- Chloroplast

- Large vacuole

- Fixed shape

Animal Cell:

- No cell wall

- No chloroplast

- Small vacuole

- Irregular shape

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16. Cell Division

Mitosis

Purpose:

Growth and repair

Produces:

2 identical daughter cells

Stages:

- Prophase

- Metaphase
- Anaphase
- Telophase

Meiosis

Purpose:

Formation of reproductive cells

Produces:

4 genetically different cells

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17. Photosynthesis

Equation:

Carbon dioxide + Water + Sunlight → Glucose + Oxygen

Occurs in:

Chloroplast

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18. Cellular Respiration

Purpose:

Energy production

Equation:

Glucose + Oxygen → Carbon dioxide + Water + ATP

Occurs in:

Mitochondria

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19. Important Terms

Cell

Basic unit of life

DNA

Genetic material

ATP

Energy currency

Organelle

Specialized structure

Photosynthesis

Food production

Respiration

Energy production

Selective permeability

Allows only certain substances

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20. Summary

Cells are the foundation of all life.

Each organelle performs a specific function.

Plant and animal cells have similarities and differences.

Understanding cells helps explain biological processes.